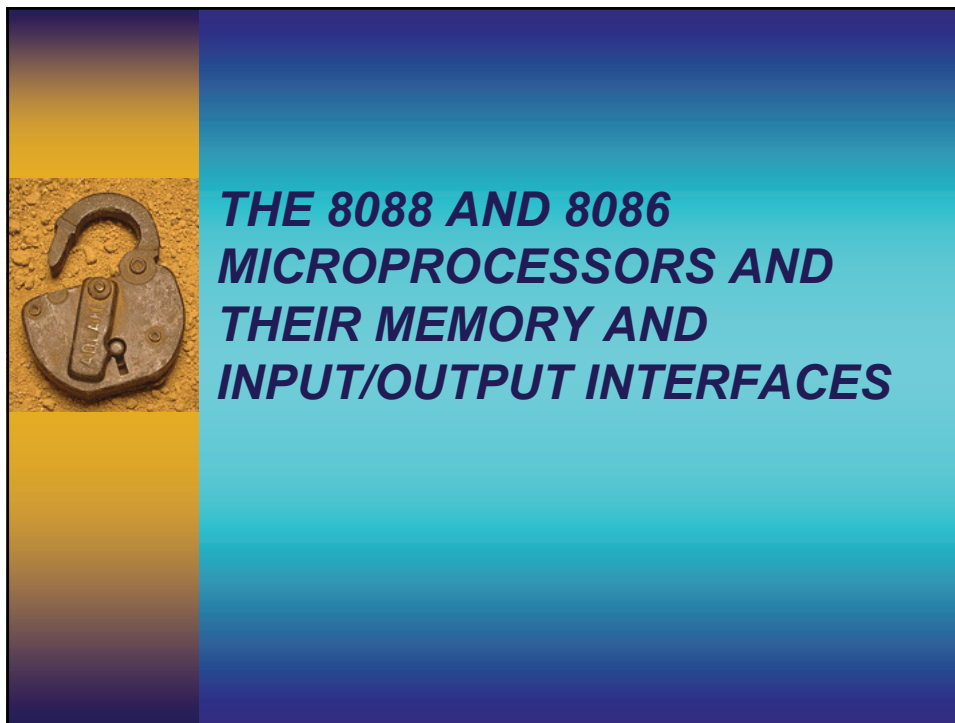


A graphic for a presentation slide. On the left, there is a vertical yellow bar. To its right, a brown metal key is shown. The background of the slide is white. The title text is in a bold, dark blue, serif font. The list of topics is in a black, sans-serif font.

***The 8088 and 8086 Microprocessors and
Their Memory and Input/output Interfaces***

- 8.1 The 8088 and 8086 Microprocessors
- 8.2 Minimum-Mode and Maximum-Mode System
- 8.3 Minimum-Mode Interface
- 8.4 Maximum-Mode Interface
- 8.5 Electrical Characteristics
- 8.6 System Clock
- 8.7 Bus Cycle and Time States
- 8.8 Hardware Organization of the Memory Address Space

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The 8088 and 8086 Microprocessors and Their Memory and Input/output Interfaces

- 8.9 Memory Bus Status Codes
- 8.10 Memory Control Signals
- 8.11 Read and Write Bus Cycles
- 8.12 Memory Interface Circuits
- 8.13 Programmable Logic Arrays
- 8.14 Types of Input/Output
- 8.15 An Isolated Input/Output Interface
- 8.16 Input/Output Data Transfer
- 8.17 Input/Output Instructions
- 8.18 Input/Output Bus Cycles

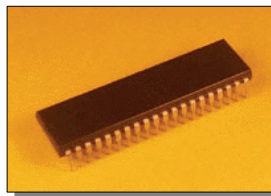
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8.1 The 8088 and 8086 Microprocessors

- The 8086, announced in 1978, was the first 16-bit microprocessor introduced by Intel Corporation.
- 8086 and 8088 are internally 16-bit MPU. However, externally the 8086 has a 16-bit data bus and the 8088 has an 8-bit data bus.



The 8088 microprocessor

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8.1 The 8088 and 8086 Microprocessors

- 8086 and 8088 both have the ability to address up to 1 Mbyte of memory and 64K of input/output port.
- The 8088 and 8086 are both manufactured using **high-performance metal-oxide semiconductor (HMOS) technology**.
- The 8088 and 8086 are housed in a 40-pin dual in-line package and many pins have multiple functions.



Intel D8086-1 Microprocessor
Processor Speed: 10.0 MHz
Bus Speed: 10.0 MHz
FPU: no
SOURCE: <http://cpu-museum.de/>

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8.1 The 8088 and 8086 Microprocessors

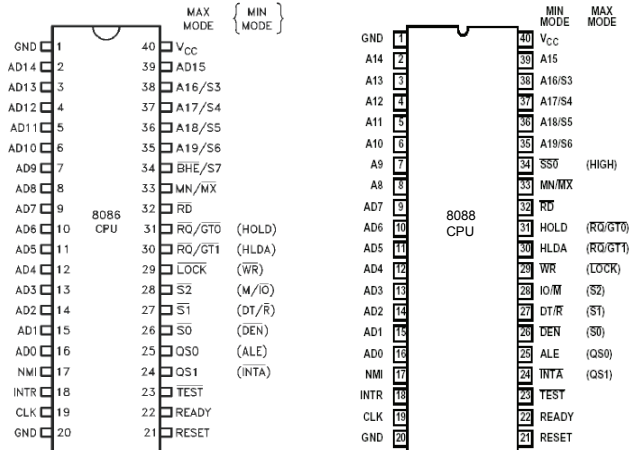
- **CMOS, Complementary Metal-Oxide-Semiconductor**, is a major class of integrated circuits used in chips such as microprocessors, microcontrollers, static RAM, digital logic circuits, and analog circuits such as image sensors.
- Two important characteristics of CMOS devices are high noise immunity and low static power supply drain. Significant power is only drawn when its transistors are switching between on and off states; consequently, CMOS devices do not produce as much heat as other forms of logic such as TTL. CMOS also allows a high density of logic functions on a chip.

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8.1 The 8088 and 8086 Microprocessors



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8.2 Minimum-Mode and Maximum-Mode System

- The 8086 and 8088 microprocessors can be configured to work in either of two modes:
 - ❖ The minimum mode - $\overline{MN}/\overline{MX} = 1$
 - ❖ The maximum mode - $\overline{MN}/\overline{MX} = 0$
- The mode selection feature lets the 8088 or 8086 better meet the needs of a wide variety of system requirement.
- Minimum mode 8088/8086 systems are typically smaller and contain a single processor.
- Depending on the mode of operation selected, the assignment for a number of the pins on the microprocessor package are changed.

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8.2 Minimum-Mode and Maximum-Mode System

Common signals		
Name	Function	Type
AD7-AD0	Address/data bus	Bidirectional, 3-state
A15-A8	Address bus	Output, 3-state
A19/S6-A16/S3	Address/status	Output, 3-state
MN/ $\overline{MX}$	Minimum/maximum Mode control	Input
$\overline{RD}$	Read control	Output, 3-state
$\overline{TEST}$	Wait on test control	Input
READY	Wait state control	Input
RESET	System reset	Input
NMI	Nonmaskable Interrupt request	Input
INTR	Interrupt request	Input
CLK	System clock	Input
V _{cc}	+5 V	Input
GND	Ground	

Signals common to both minimum and maximum mode

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8.2 Minimum-Mode and Maximum-Mode System

Minimum mode signals ($MN/\overline{MX} = V_{cc}$)		
Name	Function	Type
HOLD	Hold request	Input
HLDA	Hold acknowledge	Output
$\overline{WR}$	Write control	Output, 3-state
IO/ $\overline{M}$	IO/memory control	Output, 3-state
DT/ $\overline{R}$	Data transmit/receive	Output, 3-state
$\overline{DEN}$	Data enable	Output, 3-state
$\overline{SSO}$	Status line	Output, 3-state
ALE	Address latch enable	Output
$\overline{INTA}$	Interrupt acknowledge	Output

Unique minimum-mode signals

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8.2 Minimum-Mode and Maximum-Mode System

Maximum mode signals ($MN/\overline{MX} = GND$)		
Name	Function	Type
$\overline{RQ}/GT1, 0$	Request/grant bus access control	Bidirectional
$\overline{LOCK}$	Bus priority lock control	Output, 3-state
$\overline{S2}-\overline{S0}$	Bus cycle status	Output, 3-state
$QS1, QS0$	Instruction queue status	Output

Unique maximum-mode signals



8.2 Minimum-Mode and Maximum-Mode System

EXAMPLE

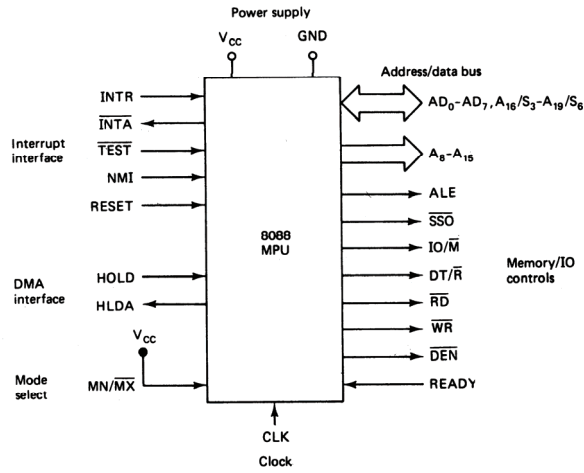
Which pins provide different signal functions in the minimum-mode 8088 and minimum-mode 8086?

Solution:

- (a) Pins 2 through 8 on the 8088 are address lines A_{14} through A_8 , but on the 8086 they are address/data lines AD_{14} through AD_8 .
- (b) Pin 28 on the 8088 is $IO/\overline{M}$ output and on the 8086 it is the $M/\overline{IO}$ output.
- (c) Pin 34 of the 8088 is the $\overline{SSO}$ output, and on the 8086 this pin supplies the $\overline{BHE}/S_7$.



8.3 Minimum-Mode Interface



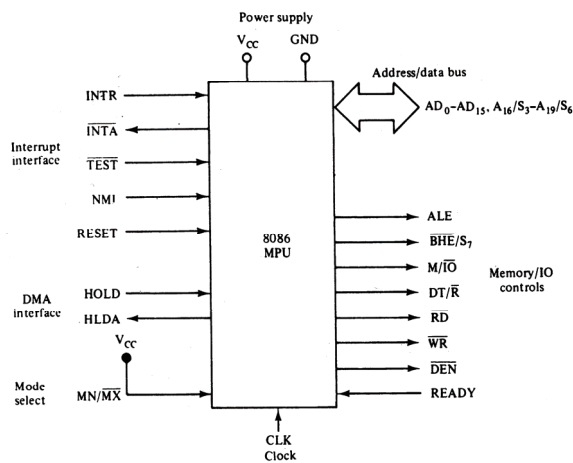
Block diagram of the minimum-mode 8088 MPU

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8.3 Minimum-Mode Interface



Block diagram of the minimum-mode 8086 MPU

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8.3 Minimum-Mode Interface

- The minimum-mode signals can be divided into the following basic groups:
 - ❖ Address/Data bus
 - ❖ Status signals
 - ❖ Control signals
 - ❖ Interrupt signals
 - ❖ DMA interface signals

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8.3 Minimum-Mode Interface

- Address/Data bus
 - ❖ The address bus is used to carry address information to the memory and I/O ports.
 - ❖ The address bus is 20-bit long and consists of signal lines A_0 through A_{19} .
 - ❖ A 20-bit address gives the 8088 a 1 Mbyte memory address space.
 - ❖ Only address line A_0 through A_{15} are used when addressing I/O. This gives an I/O address space of 64 Kbytes.
 - ❖ The 8088 has 8 multiplexed address/data bus lines ($A_0 \sim A_7$) while 8086 has 16 multiplexed address/data bus lines ($A_0 \sim A_{15}$).

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8.3 Minimum-Mode Interface

■ Status signals

- ❖ The four most significant address, A_{19} through A_{16} are multiplexed with **status signal** S_6 through S_3 .
- ❖ Bits S_4 and S_3 together form a 2-bit binary code that identifies which of the internal segment registers was used to generate the physical address. S_5 is the logic level of the internal interrupt flag. S_6 is always at the 0 logic level.

S_4	S_3	Address Status
0	0	Alternate (relative to the ES segment)
0	1	Stack (relative to the SS segment)
1	0	Code/None (relative to the CS segment or a default of zero)
1	1	Data (relative to the DS segment)

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8.3 Minimum-Mode Interface

■ Control signals

- ❖ The **control signals** are provided to support the memory and I/O interfaces of the 8088 and 8086.
 - ALE – Address Latch Enable
 - $\text{IO}/\overline{\text{M}}$ – IO/Memory (8088)
 - $\text{M}/\overline{\text{IO}}$ – Memory/IO (8086)
 - $\text{DT}/\overline{\text{R}}$ – Data Transmit/Receive (8088/8086)
 - $\overline{\text{SSO}}$ – System Status Output (8088)
 - $\overline{\text{BHE}}$ – Bank High Enable (8086)
 - $\overline{\text{RD}}$ – Read (8088/8086)
 - $\overline{\text{WR}}$ – Write (8088/8086)
 - $\overline{\text{DEN}}$ – Data Enable (8088/8086)
 - READY – Ready (8088/8086)

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8.3 Minimum-Mode Interface

■ Interrupt signals

❖ The **interrupt signals** can be used by an external device to signal that it needs to be serviced.

- INTR – Interrupt Request
- $\overline{\text{INTA}}$ – Interrupt Acknowledge
- $\overline{\text{TEST}}$ – Test (can be used to synchronize MPU)
- NMI – Nonmaskable Interrupt
- RESET – Reset (hardware reset of the MPU)

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8.3 Minimum-Mode Interface

■ DMA interface signals

- ❖ When an external device wants to take control of the system bus, it signals this fact to the MPU by switching HOLD to the 1 logic level.
- ❖ When in the hold state, signal lines AD_0 through AD_7 , A_8 through A_{15} , A_{16}/S_3 through A_{19}/S_6 , $\overline{\text{SSO}}$, IO/M , DT/R , RD , WR , $\overline{\text{DEN}}$, and INTR are all put into high-Z state.
- ❖ The 8088 signals external devices that the signal lines are in the high-Z state by switching its HLDA output to the 1 logic level.

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8.4 Maximum-Mode Interface

- The maximum-mode configuration is mainly used for implementing a **multiprocessor/coprocessor system environment**.
- **Global resources** and **local resources**
- In the maximum-mode, facilities are provided for implementing allocation of global resources and passing bus control to other microprocessors sharing the system bus.

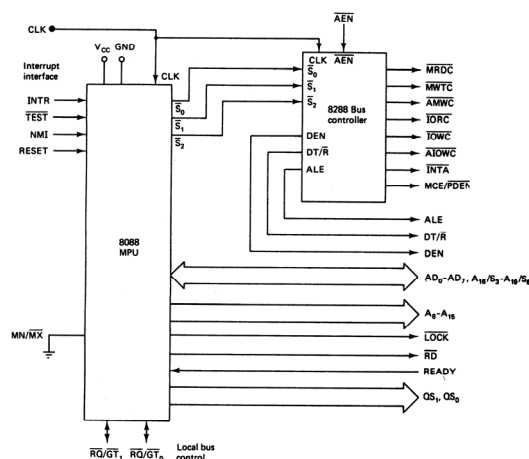
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8.4 Maximum-Mode Interface

- 8288 bus controller



8088 maximum-mode block diagram

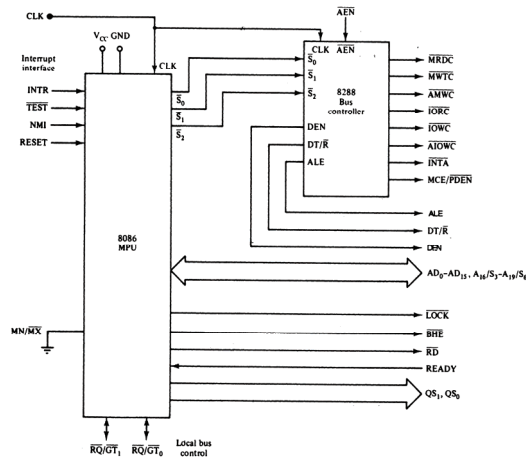
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8.4 Maximum-Mode Interface

■ 8288 bus controller



8086 maximum-mode block diagram

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8.4 Maximum-Mode Interface

■ 8288 bus controller

- ❖ In the maximum-mode, 8088/8086 outputs a status code on three signal line, S_0 , S_1 , S_2 , prior to the initialization of each bus cycle.
- ❖ The 3-bit bus status code identifies which type of bus cycle is to follow and are input to the external bus controller device, 8288.
- ❖ The 8288 produces one or two command signals for each bus cycle.

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8.4 Maximum-Mode Interface

■ 8288 bus controller

Status Inputs			CPU Cycle	8288 Command
$\overline{S_2}$	$\overline{S_1}$	$\overline{S_0}$		
0	0	0	Interrupt Acknowledge	$\overline{INTR}$
0	0	1	Read I/O Port	$\overline{IORC}$
0	1	0	Write I/O Port	$\overline{IOWC}, \overline{AIOWC}$
0	1	1	Halt	None
1	0	0	Instruction Fetch	$\overline{MRDC}$
1	0	1	Read Memory	$\overline{MRDC}$
1	1	0	Write Memory	$\overline{MWTC}, \overline{AMWC}$
1	1	1	Passive	None

Bus status code

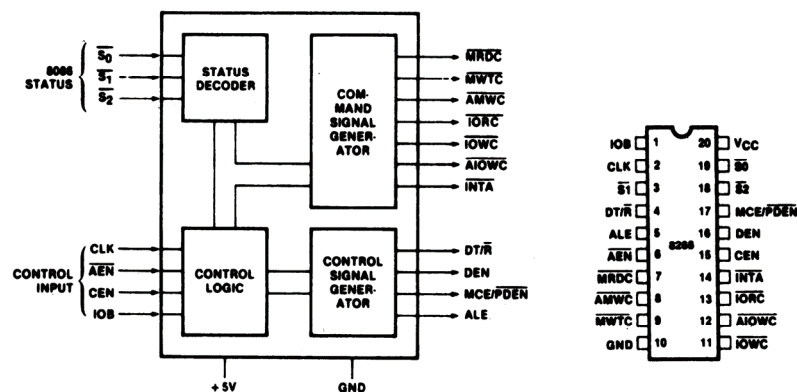
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8.4 Maximum-Mode Interface

■ 8288 bus controller



Block diagram and pin layout of 8288

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8.4 Maximum-Mode Interface

■ Lock signal

- ❖ The lock signal ($\overline{LOCK}$) is meant to be output (logic 0) whenever the processor wants to lock out the other processor from using the bus.

■ Local bus control signals

- ❖ The request/grant signals ($\overline{RQ}/\overline{GT}_0$, $\overline{RQ}/\overline{GT}_1$) provide a prioritized bus access mechanism for accessing the local bus.

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8.4 Maximum-Mode Interface

■ Queue status signals

- ❖ The 2-bit queue status code QS_0 and QS_1 tells the external circuitry what type of information was removed from the queue during the previous clock cycle.

QS1	QS0	Queue Status
0 (low)	0	No Operation. During the last clock cycle, nothing was taken from the queue.
0	1	First byte. The byte taken from the queue was the first byte of the instruction.
1 (high)	0	Queue Empty. The queue has been reinitialized as a result of the execution of a transfer of instruction.
1	1	Subsequent Byte. The byte taken from the queue was a subsequent byte of the instruction.

Queue status code

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8.4 Maximum-Mode Interface

EXAMPLE

If the bus status code $\overline{S}_2\overline{S}_1\overline{S}_0$ equals 101, what type of bus activity is taking place? Which command output is produced by the 8288?

Solution:

Looking at the bus status table, we see that bus status code 101 identifies a read memory bus cycle and causes the $\overline{MRDC}$ output of the bus controller to switch to logic 0.

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8.5 Electrical Characteristics

- Power is applied between pin 40 (V_{cc}) and pins 1 (GND) and 20 (GND).
- The nominal value of V_{cc} is specified as +5V dc with a tolerance of $\pm 10\%$.
- Both 8088 and 8086 draw a maximum of 340mA from the supply.

Symbol	Meaning	Minimum	Maximum	Test condition
V_{IL}	Input low voltage	-0.5 V	+0.8 V	
V_{IH}	Input high voltage	+2.0 V	$V_{cc} + 0.5$ V	
V_{OL}	Output low voltage		+0.45 V	$I_{OL}=2.0$ mA
V_{OH}	Output high voltage	+2.4 V		$I_{OH}=-400$ μ A

I/O voltage levels

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8.6 System Clock

- The time base for synchronization of the internal and external operations of the microprocessor in a microcomputer system is provided by the **clock** (CLK) input signal.
- The standard 8088 operates at 5 MHz and the 8088-2 operates at 8 MHz.
- The 8086 is manufactured in three speeds: 5-MHz 8086, 8-MHz 8086-2, and the 10-MHz 8086-1.
- The CLK is externally generated by the 8284 clock generator and driver IC.

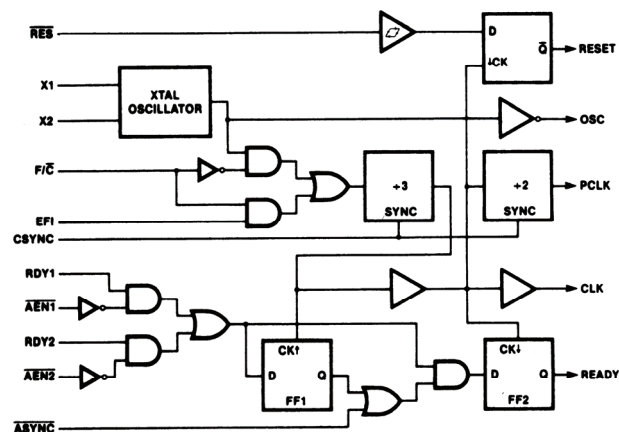
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8.6 System Clock

- Block diagram of the 8284 clock generator



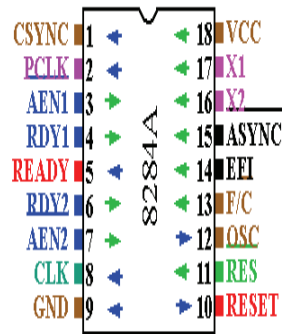
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8.6 System Clock

- Block diagram of the 8284 clock generator



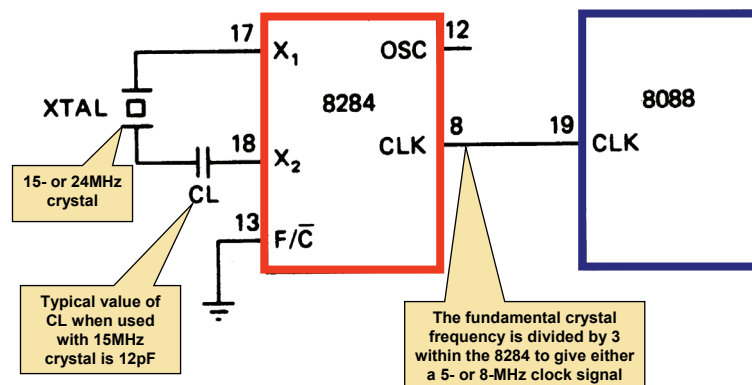
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8.6 System Clock

- Connecting the 8284 to the 8088



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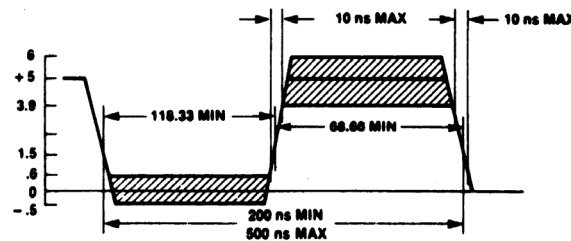
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8.6 System Clock

■ CLK waveform

- ❖ The signal is specified at Metal Oxide Semiconductor (MOS)-compatible voltage level.
- ❖ The period of the 5-MHz 8088 can range from 200 ns to 500 ns, and the maximum rise and fall times of its edges equal 10 ns.



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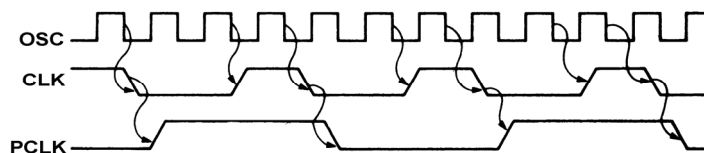
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8.6 System Clock

■ PCLK and OSC signals

- ❖ The peripheral clock (PCLK) and oscillator clock (OSC) signals are provided to drive peripheral ICs.
- ❖ The clock output at PCLK is half the frequency of CLK. The OSC output is at the crystal frequency which is three times of CLK.



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8.6 System Clock

EXAMPLE

If the CLK input of an 8086 MPU is to be driven by a 9-MHz signal, what speed version of the 8086 must be used and what frequency crystal must be attached to the 8284

Solution:

The 8086-1 is the version of the 8086 that can be run at 9-MHz. To create the 9-MHz clock, a 27-MHz crystal must be used on the 8284.

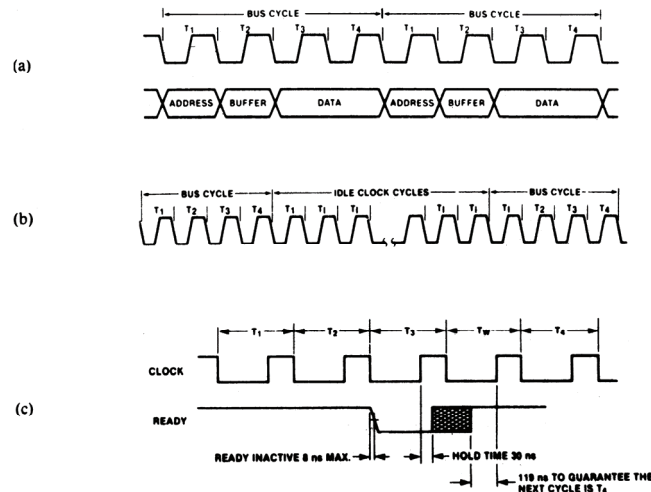


8.7 Bus Cycle and Time States

- A bus cycle defines the basic operation that a microprocessor performs to communicate with external devices.
- Examples of bus cycles are the memory read, memory write, input/output read, and input/output write.
- The bus cycle of the 8088 and 8086 microprocessors consists of at least **four clock periods**.
- If no bus cycles are required, the microprocessor performs what are known as **idle states**.
- When READY is held at the 0 level, **wait states** are inserted between states T_3 and T_4 of the bus cycle.



8.7 Bus Cycle and Time States



Bus cycle clock periods, idle state, and wait state

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8.7 Bus Cycle and Time States

EXAMPLE

What is the duration of the bus cycle in the 8088-based microcomputer if the clock is 8 MHz and the two wait states are inserted.

Solution:

The duration of the bus cycle in an 8 MHz system is given by

$$t_{cyc} = 500 \text{ ns} + N \times 125 \text{ ns}$$

In this expression the N stands for the number of wait states. For a bus cycle with two wait states, we get

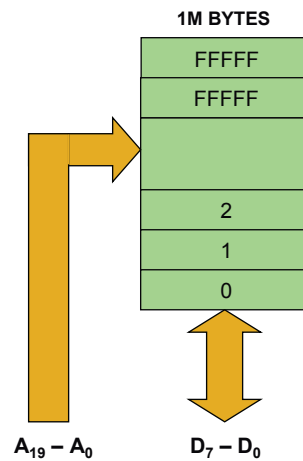
$$t_{cyc} = 500 \text{ ns} + 2 \times 125 \text{ ns} = 500 \text{ ns} + 250 \text{ ns} = 750 \text{ ns}$$

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8.8 Hardware Organization of the Memory Address Space



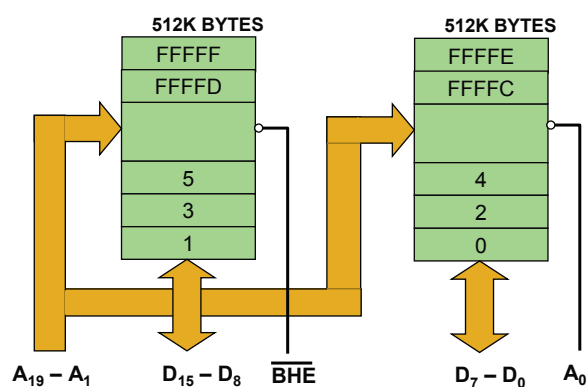
1Mx8 memory bank of the 8088

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8.8 Hardware Organization of the Memory Address Space



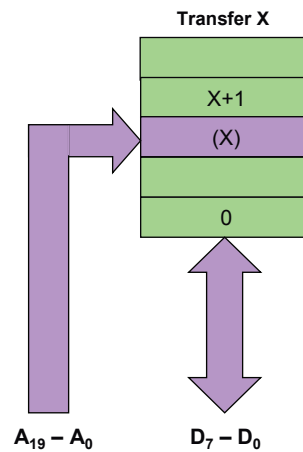
High and low memory banks of the 8086

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8.8 Hardware Organization of the Memory Address Space



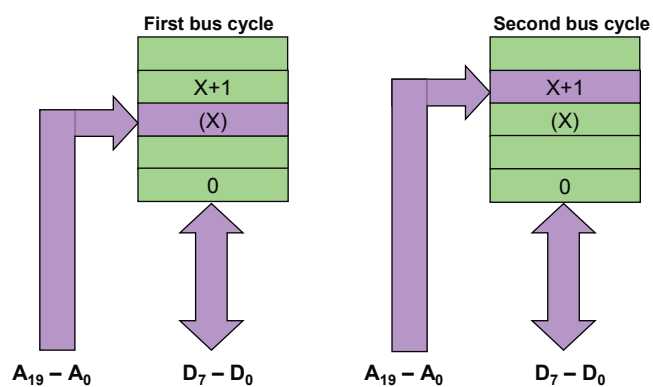
Byte transfer by the 8088

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8.8 Hardware Organization of the Memory Address Space



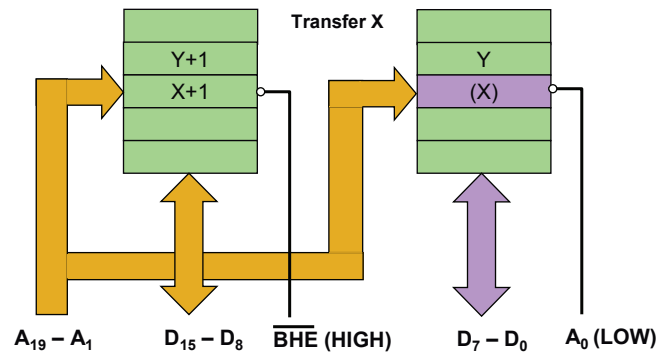
Word transfer by the 8088

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8.8 Hardware Organization of the Memory Address Space



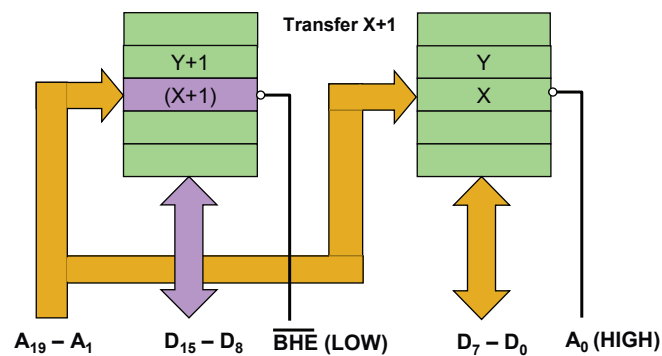
Even address byte transfer by the 8086

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8.8 Hardware Organization of the Memory Address Space



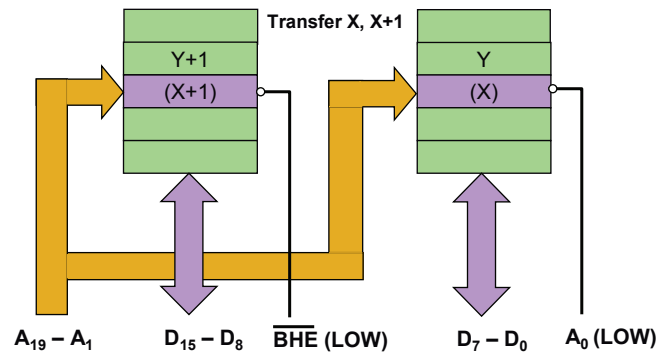
Odd address byte transfer by the 8086

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8.8 Hardware Organization of the Memory Address Space



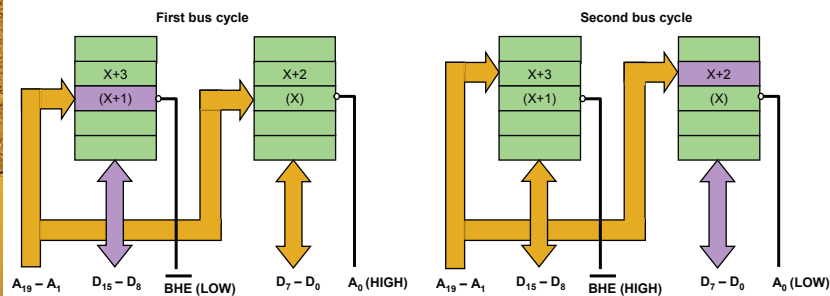
Even address word transfer by the 8086

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8.8 Hardware Organization of the Memory Address Space



Odd-address word transfer by the 8086

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8.8 Hardware Organization of the Memory Address Space

EXAMPLE

Is the word at memory address 01231_{16} of an 8086-based microcomputer aligned or misaligned? How many cycle are required to read it from memory?

Solution:

The first byte of the word is the second byte at the aligned-word address 01230_{16} . Therefore, the word is misaligned and required two bus cycles to be read from memory.

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8.9 Address Bus Status Codes

- Whenever a memory bus cycle is in progress, an address bus status code S_4S_3 is output by the processor.
- S_4S_3 identifies which one of the four segment register is used to generate the physical address in the current bus cycle:
 - ❖ $S_4S_3=00$ identifies the extra segment register (ES)
 - ❖ $S_4S_3=01$ identifies the stack segment register (SS)
 - ❖ $S_4S_3=10$ identifies the code segment register (CS)
 - ❖ $S_4S_3=11$ identifies the data segment register (DS)
- The memory address reach of the microprocessor can thus be expanded to 4 Mbytes.

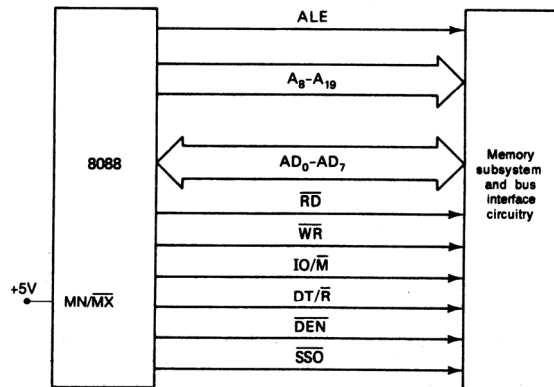
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8.10 Memory Control Signals

■ Minimum-mode memory control signals



Minimum-mode 8088 memory interface

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8.10 Memory Control Signals

■ Minimum-mode memory control signals (8088)

- ❖ **ALE – Address Latch Enable** – used to latch the address in external memory.
- ❖ **IO/ $\overline{M}$ – Input-Output/Memory** – signal external circuitry whether a memory or I/O bus cycle is in progress.
- ❖ **DT/ $\overline{R}$ – Data Transmit/Receive** – signal external circuitry whether the 8088 is transmitting or receiving data over the bus.
- ❖ **$\overline{RD}$ – Read** – identifies that a read bus cycle is in progress.
- ❖ **$\overline{WR}$ – Write** – identifies that a write bus cycle is in progress.
- ❖ **$\overline{DEN}$ – Data Enable** – used to enable the data bus.
- ❖ **$\overline{SSO}$ – Status Line** – identifies whether a code or data access is in progress.

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8.10 Memory Control Signals

- The control signals for the 8086's minimum-mode memory interface differs in three ways:
 - ❖ $\overline{IO}/\overline{M}$ signal is replaced by $\overline{M}/\overline{IO}$ signal.
 - ❖ The signal $\overline{SSO}$ is removed from the interface.
 - ❖ $\overline{BHE}$ (bank high enable) is added to the interface and is used to select input for the high bank of memory in the 8086's memory subsystem.

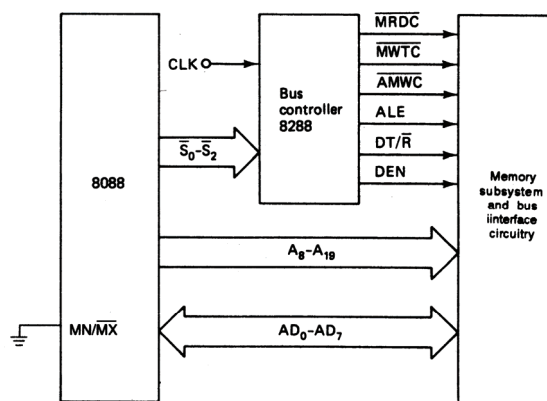
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8.10 Memory Control Signals

- Maximum-mode memory control signals



Maximum-mode 8088 memory interface

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8.10 Memory Control Signals

Maximum-mode memory control signals

- ❖ $\overline{\text{MRDC}}$ – Memory Read Command
- ❖ $\overline{\text{MWTC}}$ – Memory Write Command
- ❖ $\overline{\text{AMWC}}$ – Advanced Memory Write Command

Status Inputs			CPU Cycle	8288 Command
$\overline{\text{S}}_2$	$\overline{\text{S}}_1$	$\overline{\text{S}}_0$		
0	0	0	Interrupt Acknowledge	$\overline{\text{INTA}}$
0	0	1	Read I/O Port	$\overline{\text{IORC}}$
0	1	0	Write I/O Port	$\overline{\text{IOWC}}, \overline{\text{AIOWC}}$
0	1	1	Halt	None
1	0	0	Instruction Fetch	$\overline{\text{MRDC}}$
1	0	1	Read Memory	$\overline{\text{MRDC}}$
1	1	0	Write Memory	$\overline{\text{MWTC}}, \overline{\text{AMWC}}$
1	1	1	Passive	None

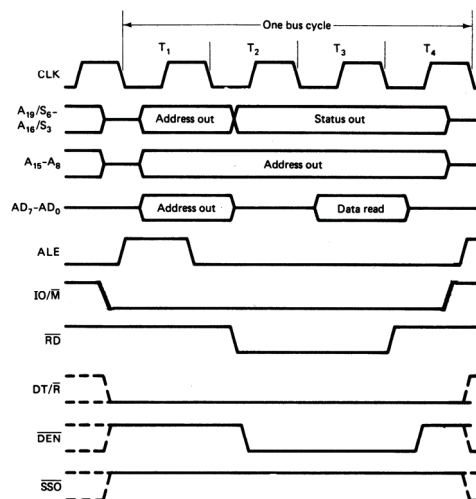
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8.11 Read and Write Bus Cycle

Read cycle



Minimum-mode memory read bus cycle of the 8088

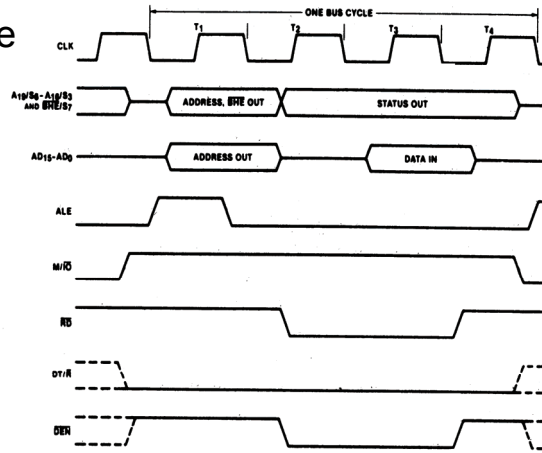
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8.11 Read and Write Bus Cycle

■ Read cycle



Minimum-mode memory read bus cycle of the 8086

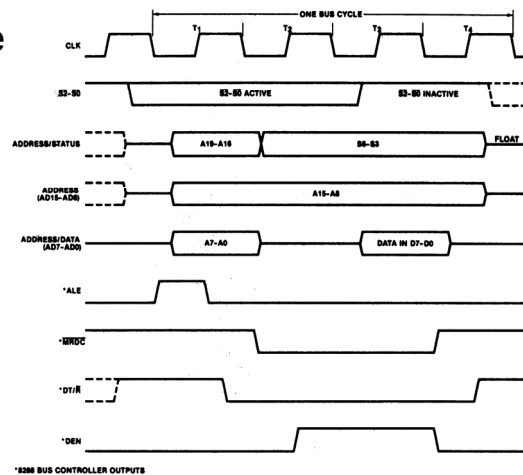
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8.11 Read and Write Bus Cycle

■ Read cycle



Maximum-mode memory read bus cycle of the 8086

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A close-up photograph of a single, dark, weathered metal key. The key has a simple, rounded bow at the top and a bit with several notches. It is resting on a textured, reddish-brown surface, possibly a piece of wood or stone. The lighting is warm, highlighting the texture of the metal and the surface it lies on.



Mi

61.

國
生



A large, ornate metal key with a circular bow and a notched bit, resting on a textured, reddish-brown surface.

M:

61.

國
生



8.12 Memory Interface Circuit

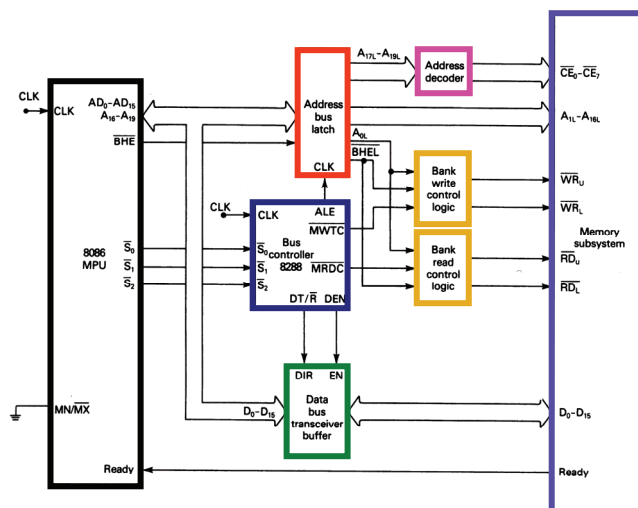
- Address bus latches and buffers
- Bank write and bank read control logic
- Data bus transceivers/buffers
- Address decoders

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8.12 Memory Interface Circuit



Memory interface block diagram

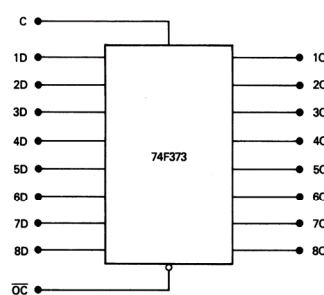
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8.12 Memory Interface Circuit

■ Address bus latches and buffers



Operation of the 74F373

Inputs			Output
$\overline{OC}$	Enable C	D	Q
L	H	H	H
L	H	L	L
L	L	X	Q_0
H	X	X	Z

Block diagram of a D-type latch

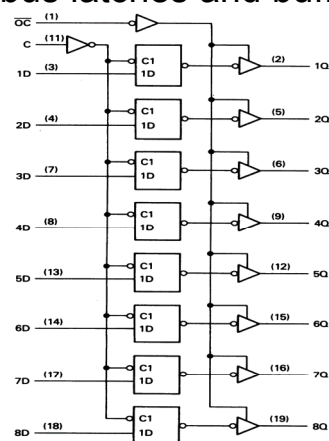
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8.12 Memory Interface Circuit

■ Address bus latches and buffers



Circuit diagram of the 74F373

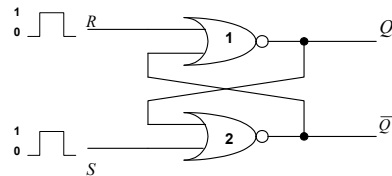
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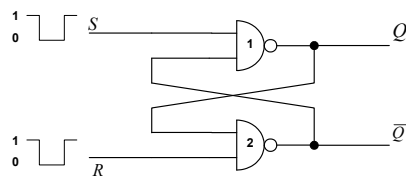


8.12 Memory Interface Circuit

■ A review of flip-flop/latch logic



Cross-NOR S-R flip-flop



Cross-NAND S-R flip-flop

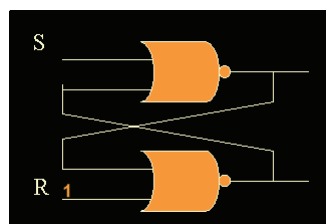
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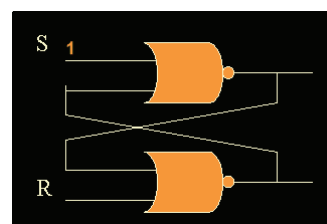


8.12 Memory Interface Circuit

■ A review of flip-flop/latch logic



RESET



SET

Cross-NOR S-R flip-flop

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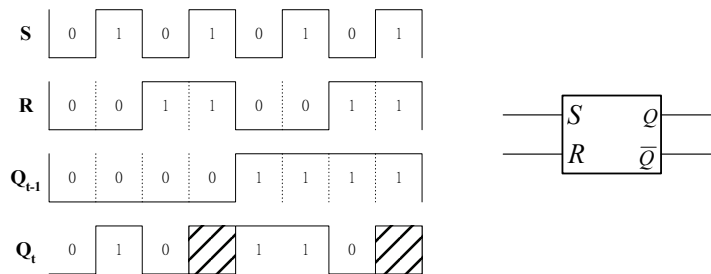
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8.12 Memory Interface Circuit

■ A review of flip-flop/latch logic

The D latch is used to capture, or 'latch' the logic level which is present on the data line when the clock input is high.



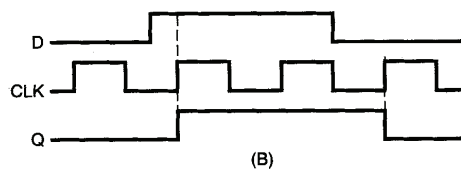
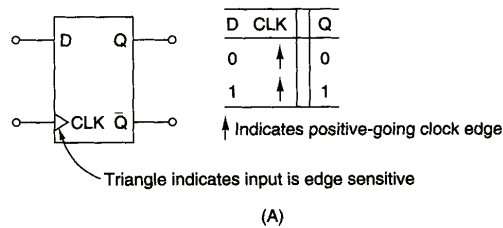
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8.12 Memory Interface Circuit

■ A review of flip-flop/latch logic



Positive edge-triggered D flip-flop

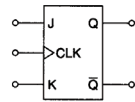
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8.12 Memory Interface Circuit

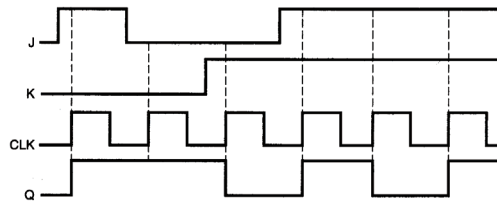
■ A review of flip-flop/latch logic



J	K	CLK	Q
0	0	↑	Q_0 (no change)
1	0	↑	1
0	1	↑	0
1	1	↑	$\bar{Q}_0$ (toggles)

Q_0 is state Q just prior to ↑ of CLK.

(A)



(B)

Positive edge-triggered JK flip-flop

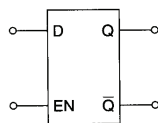
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8.12 Memory Interface Circuit

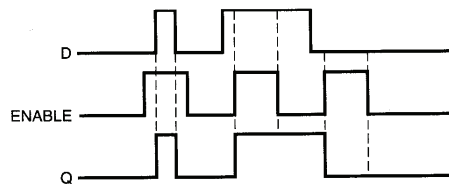
■ A review of flip-flop/latch logic



Inputs		Output
EN	D	Q
0	X	Q_0 (no change)
1	0	0
1	1	1

"X" indicates "don't care"
 Q_0 is state Q just prior to EN going LOW

(A)



(B)

D-type latch

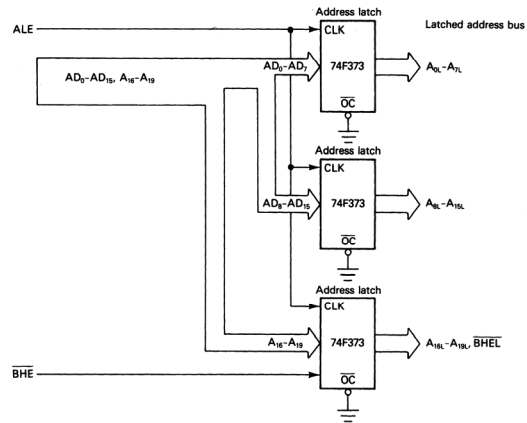
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8.12 Memory Interface Circuit

■ Address bus latches and buffers



Address latch circuit

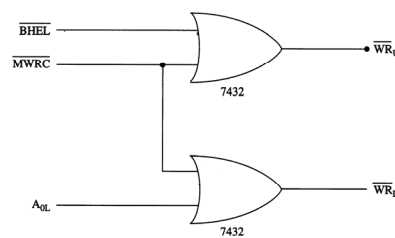
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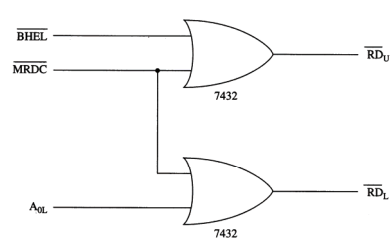


8.12 Memory Interface Circuit

■ Bank write and bank read control logic



Bank write control logic



Bank read control logic

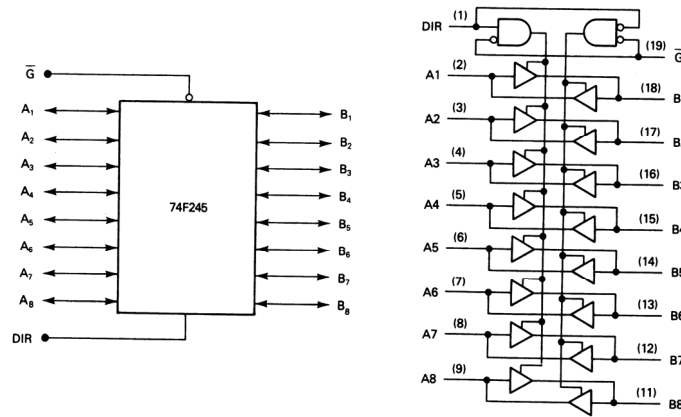
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8.12 Memory Interface Circuit

■ Data bus transceivers



Block diagram and circuit diagram of the 74F245 octal bus transceiver

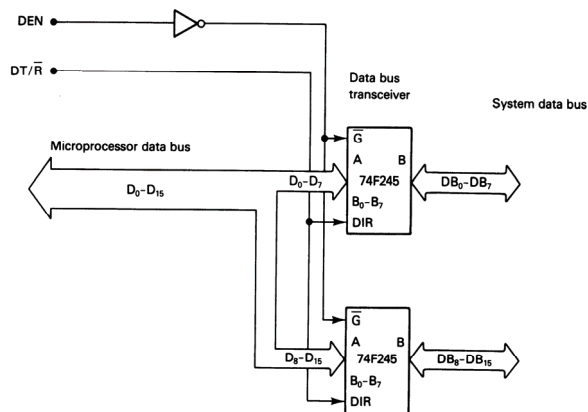
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8.12 Memory Interface Circuit

■ Data bus transceivers



Data bus transceiver circuit

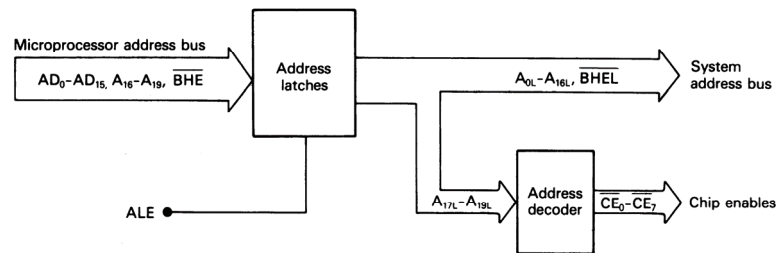
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8.12 Memory Interface Circuit

■ Address decoder

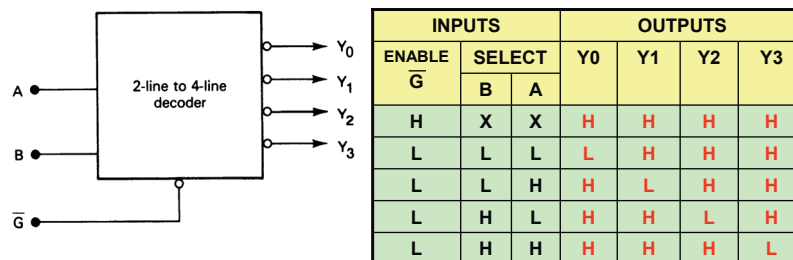


Address bus configuration with address decoding



8.12 Memory Interface Circuit

■ Address decoder

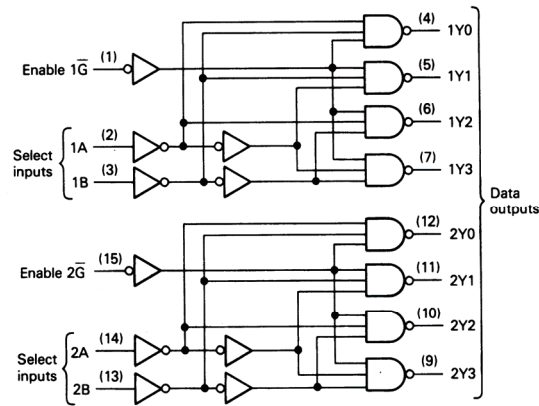


Block diagram and operation of the 74F139 decoder



8.12 Memory Interface Circuit

■ Address decoder



Circuit diagram of the 74F139 decoder

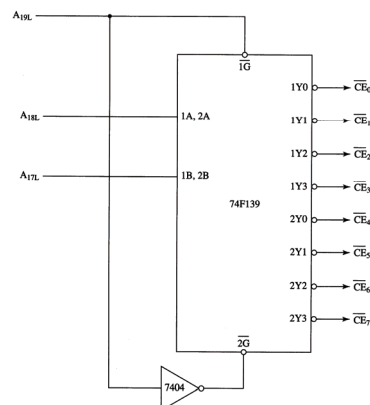
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8.12 Memory Interface Circuit

■ Address decoder



Address decoder circuit using 74F139

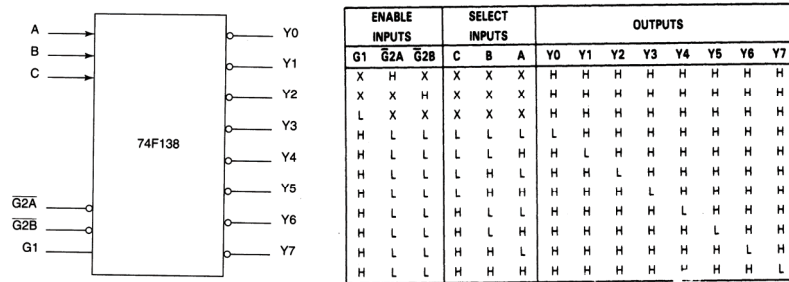
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8.12 Memory Interface Circuit

■ Address decoder



Block diagram and operation of the 74F138 decoder

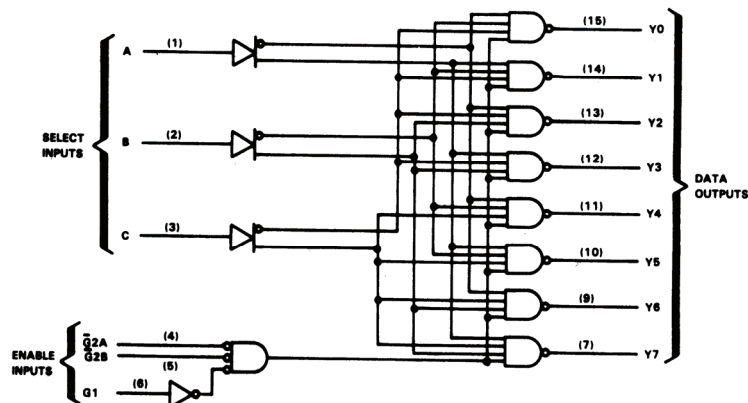
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8.12 Memory Interface Circuit

■ Address decoder



Circuit diagram of the 74F138 decoder

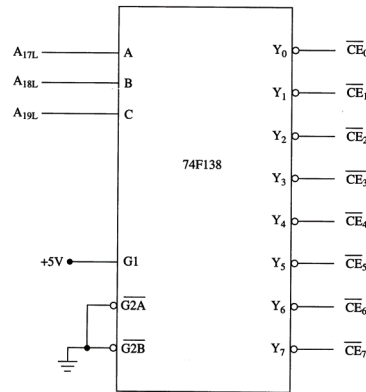
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8.12 Memory Interface Circuit

■ Address decoder



Address decoder circuit using 74F138

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8.13 Programmable Logic Arrays

- **Programmable logic array, PLA**, are general-purpose logic devices that have the ability to perform a wide variety of specialized logic functions.
- A PLA contains a general-purpose AND-OR-NOT array of logic gate circuits.
- The process used to connect or disconnect inputs of the AND gate array is known as programming, which leads to the name programmable logic array.

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8.13 Programmable Logic Arrays

■ Major types of programmable logic architecture

- ❖ Simple Programmable Logic Devices (**SPLDs**)
 - PAL, GAL, PLA, EPLD
- ❖ Complex Programmable Logic Devices (**CPLDs**)
 - EPLD, PEEL, EEPLD, MAX
- ❖ Field Programmable Gate Arrays (**FPGAs**)
 - LCA, pASIC, FLEX, APEX, ACT, ORCA, Virtex, pASIC
- ❖ Field Programmable InterConnect (**FPICs**)



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8.13 Programmable Logic Arrays

■ PLAs, GALs, and EPLDs

- ❖ Early PLA devices were all manufactured with the bipolar semiconductor process.
- ❖ Bipolar devices are programmed with an interconnect pattern by burning out fuse links within the device.
- ❖ PLAs made with bipolar technology are characterized by slower operating speeds and higher power consumption.
- ❖ Two kinds of newer PLA, manufactured with the CMOS process, are in wide use today: the **GAL** and **EPLD**.

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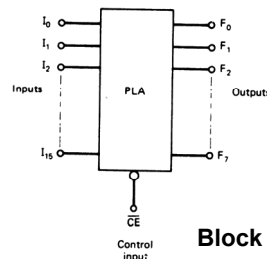
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8.13 Programmable Logic Arrays

■ Block diagram of a PLA

- ❖ The logic levels applied at inputs I_0 through I_{15} and the programming of the AND array determine what logic levels are produced at outputs F_0 through F_{15} .
- ❖ The capacity of a PLA is measured by three properties: the number of inputs, the number of outputs, and the number of product terms (P-terms)



Block diagram of a PLA

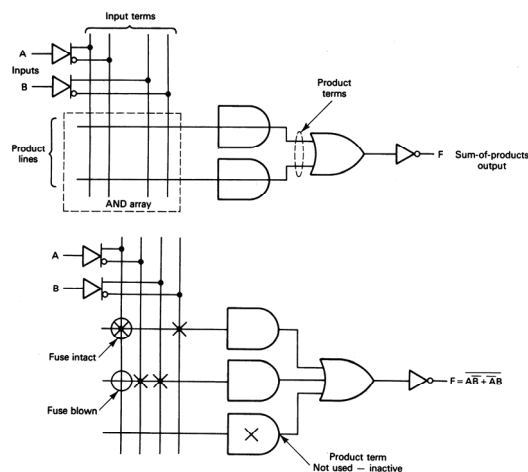
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8.13 Programmable Logic Arrays

■ Architecture of a PLA



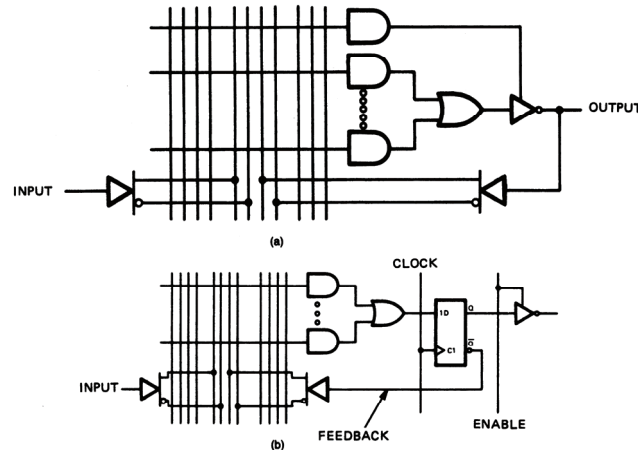
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8.13 Programmable Logic Arrays

■ Architecture of a PLA



(a) Typical PLA architecture. (b) PLA with output latch

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8.13 Programmable Logic Arrays

■ Standard PAL™ device

- ❖ A PAL, programmable array logic, is a PLA in which the OR array is fixed; only the AND array is programmable.
- ❖ The 16L8 is a widely used PAL IC. It is housed in a 20-pin package. It has 10 dedicated input, 2 dedicated outputs, and 6 programmable I/O lines.
- ❖ The 16L8 is manufactured with bipolar technology. It operates from a $+5V \pm 10\%$ dc power supply and draw a maximum of 180mA.
- ❖ The 20L8 has 20 inputs, 8 outputs and 64 P-terms.
- ❖ The 20R8 is the register output version of 20L8.

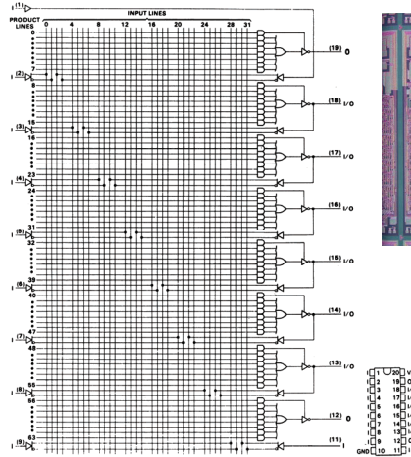
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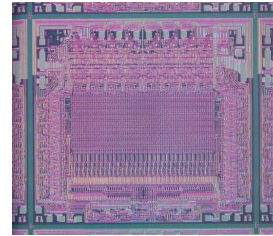
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8.13 Programmable Logic Arrays

■ Standard PAL™ device



16L8 circuit diagram and pin layout



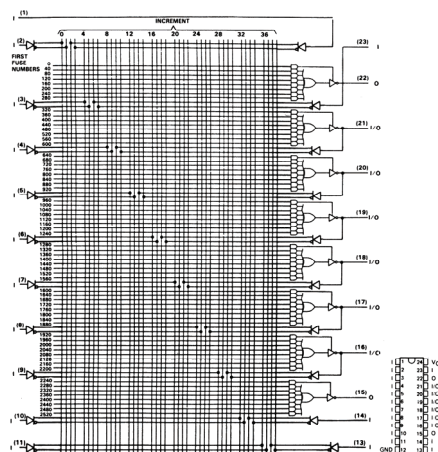
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8.13 Programmable Logic Arrays

■ Standard PAL™ device



20L8 circuit diagram and pin layout

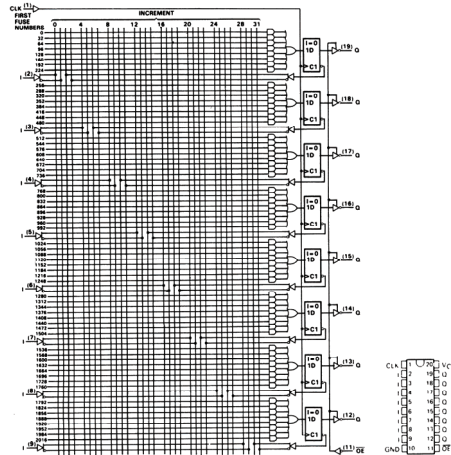
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8.13 Programmable Logic Arrays

■ Standard PAL™ device



16R8 circuit diagram and pin layout

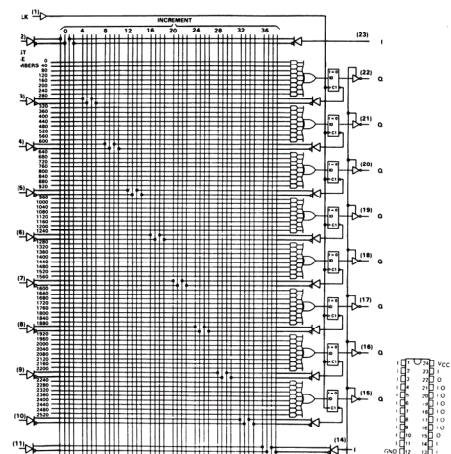
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8.13 Programmable Logic Arrays

■ Standard PAL™ device



20R8 circuit diagram and pin layout

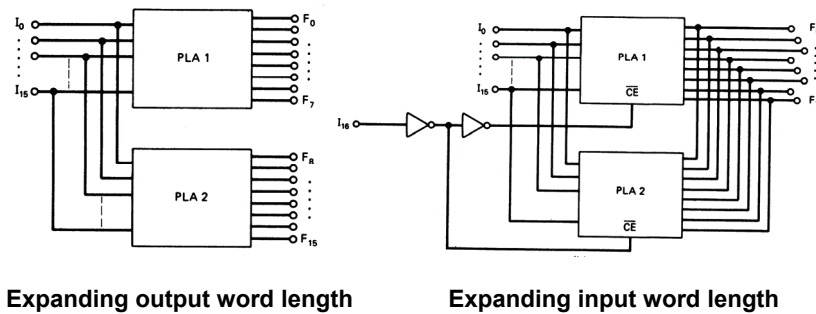
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8.13 Programmable Logic Arrays

■ Expanding PLA capacity



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8.14 Types of Input/Output

■ Isolated input/output

- ❖ When using isolated I/O in a microcomputer system, the I/O device are treated separate from memory.
- ❖ The memory address space contains 1 M consecutive byte address in the range 00000_{16} through $FFFFF_{16}$; and that the I/O address space contains 64K consecutive byte addresses in the range 0000_{16} through $FFFF_{16}$.
- ❖ All input and output data transfers must take place between the AL or AX register and I/O port.

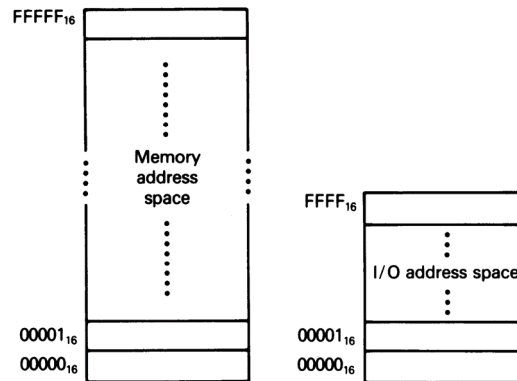
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8.14 Types of Input/Output

■ Isolated input/output



8088/8086 memory and I/O address spaces

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8.14 Types of Input/Output

■ Memory-mapped input/output

- ❖ In the case of memory-mapped I/O, MPU looks at the I/O port as though it is a storage location in memory.
- ❖ Some of the memory address space is dedicated to I/O ports.
- ❖ Instructions that affect data in memory are used instead of the special I/O instructions.
- ❖ The memory instructions tend to execute slower than those specifically designed for isolated I/O.

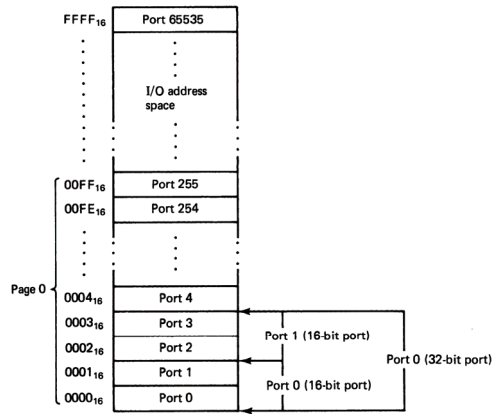
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8.14 Types of Input/Output

■ Memory-mapped input/output



Isolated I/O ports

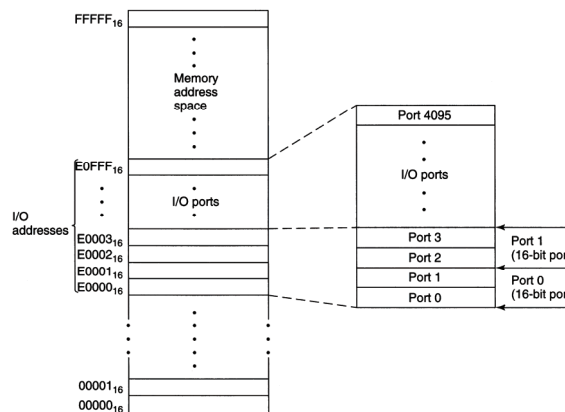
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8.14 Types of Input/Output

■ Memory-mapped input/output



Memory mapped I/O ports

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8.15 Isolated Input/Output Interface

■ I/O devices:

- ❖ Keyboard
- ❖ Printer
- ❖ Mouse
- ❖ 82C55A, etc.

■ Functions of interface circuit:

- ❖ Select the I/O port
- ❖ Latch output data
- ❖ Sample input data
- ❖ Synchronize data transfer
- ❖ Translate between TTL voltage levels and those required to operate the I/O devices.

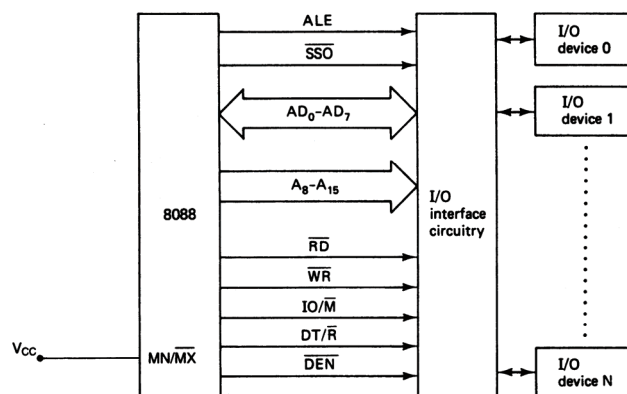
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8.15 Isolated Input/Output Interface

■ Minimum-mode interface



Minimum-mode 8088 system I/O interface

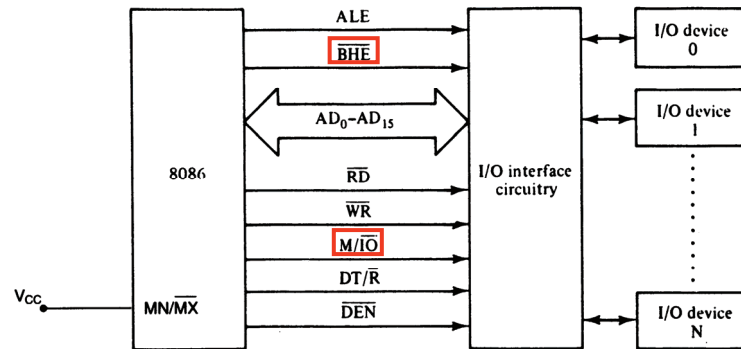
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8.15 Isolated Input/Output Interface

■ Minimum-mode interface



Minimum-mode 8086 system I/O interface

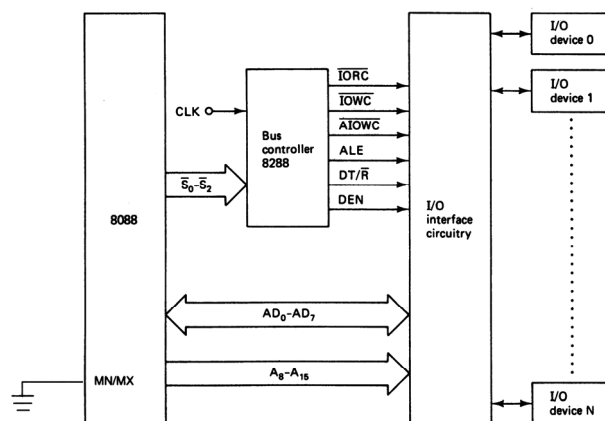
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8.15 Isolated Input/Output Interface

■ Maximum-mode interface



Maximum-mode 8088 system I/O interface

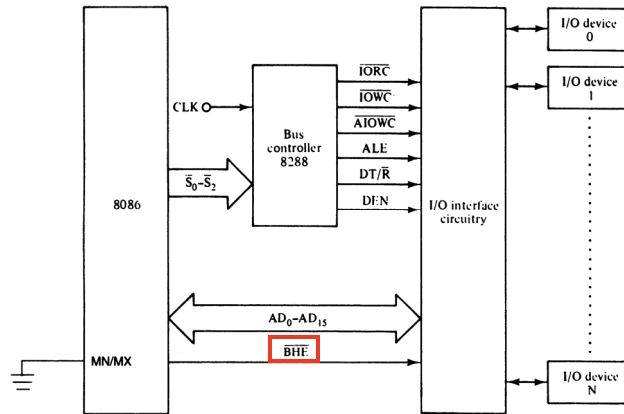
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8.15 Isolated Input/Output Interface

■ Maximum-mode interface



Maximum-mode 8086 system I/O interface

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8.15 Isolated Input/Output Interface

■ Maximum-mode interface

Status Inputs			CPU Cycle	8288 Command
$\overline{S_2}$	$\overline{S_1}$	$\overline{S_0}$		
0	0	0	Interrupt Acknowledge	$\overline{INTA}$
0	0	1	Read I/O Port	$\overline{IORC}$
0	1	0	Write I/O Port	$\overline{IOWC}, \overline{AIOWC}$
0	1	1	Halt	None
1	0	0	Instruction Fetch	$\overline{MRDC}$
1	0	1	Read Memory	$\overline{MRDC}$
1	1	0	Write Memory	$\overline{MWTC}, \overline{AMWC}$
1	1	1	Passive	None

I/O bus cycle status codes

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8.16 Input/Output Data Transfers

- Input/output data transfers in the 8088 and 8086 microcomputers can be either byte-wide or word-wide.
- I/O addresses are 16 bits in length and are output by the 8088 to the I/O interface over bus lines AD_0 through AD_7 and A_8 through A_{15} .
- In 8088, the word transfers is performed as two consecutive byte-wide data transfer and takes two bus cycle.
- In 8086, the word transfers can takes either one or two bus cycle.
- Word-wide I/O ports should be aligned at even-address boundaries.

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8.17 Input/Output Instructions

Mnemonic	Meaning	Format	Operation
IN	Input direct	IN Acc, Port	$(Acc) \leftarrow (Port)$ Acc = AL or AX
	Input indirect (variable)	IN Acc, DX	$(Acc) \leftarrow ((DX))$
OUT	Output direct	OUT Port, Acc	$(Port) \leftarrow (Acc)$
	Output indirect (variable)	OUT DX, Acc	$((DX)) \leftarrow (Acc)$

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8.17 Input/Output Instructions

EXAMPLE

Write a sequence of instructions that will output the data FF_{16} to a byte-wide output port at address AB_{16} of the I/O address space.

Solution:

First, the AL register is loaded with FF_{16} as an immediate operand in the instruction

```
MOV AL, 0FFH
```

Now the data in AL can be output to the byte-wide output port with the instruction

```
OUT 0ABH, AL
```



8.17 Input/Output Instructions

EXAMPLE

Write a series of instructions that will output FF_{16} to an output port located at address $B000_{16}$ of the I/O address space.

Solution:

The DX register must first be loaded with the address of the output port. This is done with the instruction

```
MOV DX, 0B000H
```

Next, the data that are to be output must be loaded into AL with the instruction

```
MOV AL, 0FFH
```

Finally, the data are output with the instruction

```
OUT DX, AL
```



8.17 Input/Output Instructions

EXAMPLE

Data are to be read in from two byte-wide input ports at addresses AA_{16} and $A9_{16}$ and then output as a word-wide output port at address $B000_{16}$. Write a sequence of instructions to perform this input/output operation.

Solution:

First read in the byte at address AA_{16} into AL and move it into AH.

```
IN    AL, 0AAH
MOV   AH, AL
```

Now the other byte can be read into AL by the instruction

```
IN    AL, 09AH
```

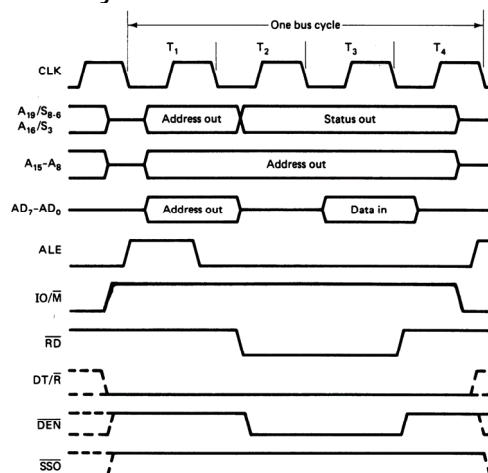
And to write out the word of data

```
MOV   DX, 0B000H
OUT   DX, AX
```



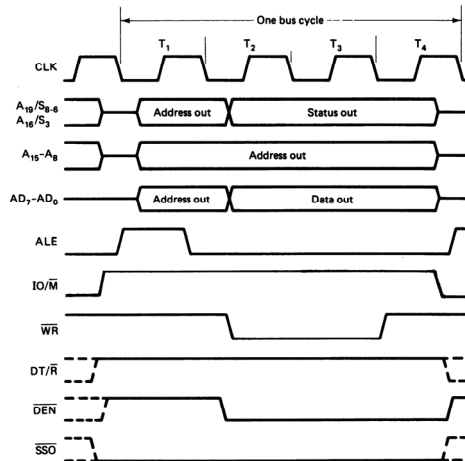
8.18 Input/Output Bus Cycle

■ Input bus cycle of the 8088



8.18 Input/Output Bus Cycle

■ Output bus cycle of the 8088



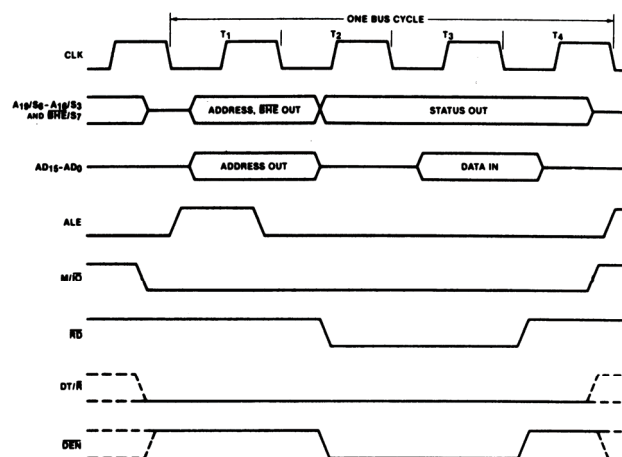
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8.18 Input/Output Bus Cycle

■ Input bus cycle of the 8086



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8.18 Input/Output Bus Cycle

■ Output bus cycle of the 8086

